GOVERNMENT ENGINEERING COLLEGE BARTON HILL

ECE DEPARTMENT NEWSLETTER

EDITION #17

EDITORIAL TEAM:
KEDAR KAMALRAJ, S5 ECE
NIRANJAN B NAIR, S3 ECE
ABHILASH A S, S3 ECE

FACULTY COORDINATOR:
Prof. Neethu P R

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VISION

A learning hub in electronics and communication engineering, that enables students develop engineering fundamentals, and that builds bidirectional bridges with leading industries and world-class research centers.

MISSION

(**M1**) Establish a learning environment that enables students build the fundamentals and study advanced disciplines related to electronics and communication engineering.

(**M2**) Promote innovation and aptitude for research by exposing students and faculty into new areas of technology.

(**M3**) Provide an ethical and value-based education by encouraging activities addressing societal needs.

EDITOR'S NOTE

Dear Readers,

It is with great excitement that we bring you this edition of the ECE Department Newsletter, celebrating the dynamic world of Electronics and Communication Engineering. As one of the most transformative fields in modern technology, ECE continues to shape industries, drive innovation, and improve everyday life through groundbreaking advancements in communication, embedded systems, artificial intelligence, and more.

Our students and faculty have been actively contributing to research, technical events, and professional organizations like IEEE, further strengthening our department's legacy of excellence. Their passion, dedication, and curiosity fuel the technological progress we witness today and inspire the innovations of tomorrow.

This newsletter serves as a platform to showcase achievements, explore new ideas, and highlight the remarkable potential of ECE. We encourage students to seize every opportunity to learn, innovate, and contribute to the ever-evolving world of electronics and communication.

Wishing you all success in your academic and professional journeys!

Warm regards,
Editorial Team,
ECE Department Newsletter
Government Engineering College Barton Hill

HOD's Note



Dr. Sanil K Daniel
HEAD OF THE DEPARTMENT

Dear Students, Faculty, and Esteemed Stakeholders,

It is with great pride and enthusiasm that I extend my warm greetings to everyone associated with the Electronics and Communication Engineering (ECE) Department at Government Engineering College, Barton Hill.

Our department has always been dedicated to academic excellence, innovation, and research in the ever-evolving field of Electronics and Communication Engineering. Today, as we witness rapid advancements in Artificial Intelligence, the Internet of Things (IoT), 6G, VLSI, and Embedded Systems, the role of an Electronics and Communication Engineer has become more crucial than ever. From smart cities and autonomous vehicles to breakthrough developments in healthcare and defense, our expertise has a profound impact on the world around us.

We are committed to equipping our students with the technical knowledge, problem-solving skills, and ethical foundation necessary to meet real-world challenges. But education is not just about mastering concepts—it is about applying them to create meaningful solutions. I encourage you to cultivate curiosity, embrace innovation, and actively engage in research and entrepreneurship. The opportunities in emerging technologies are vast, and your passion and dedication will shape the future of this field.

As you move forward in your academic and professional journeys, remember that integrity, perseverance, and a lifelong learning mindset will be your greatest assets. Be open to change, adapt to new challenges, and strive to be pioneers in your respective domains.

Wishing you success and fulfilment in all your endeavours!

Dr. Sanil K Daniel
Head of the Department
Electronics and Communication Engineering
Government Engineering College, Barton Hill

Famous Personalities Who Contributed to ECE

Electronics and Communication Engineering (ECE) has been shaped by brilliant minds whose groundbreaking contributions have transformed technology and society. Here are some of the most influential personalities in the field:

Claude Shannon (1916-2001) – Known as the "Father of Information Theory," Shannon laid the foundation for digital communication and data compression, which form the backbone of modern communication systems.

Nikola Tesla (1856-1943) – A visionary inventor, Tesla made significant contributions to alternating current (AC) power systems, wireless communication, and radio waves, paving the way for modern electronics.

John Bardeen (1908-1991) – A two-time Nobel Prize winner in physics, Bardeen co-invented the transistor, which revolutionized electronic circuits and led to the development of modern computing.

William Shockley (1910-1989) & Walter Brattain (1902-1987) – Along with John Bardeen, they co-invented the transistor, which replaced bulky vacuum tubes and transformed the electronics industry.

Jack Kilby (1923-2005) & Robert Noyce (1927-1990) – Kilby and Noyce independently developed the integrated circuit (IC), leading to the miniaturization of electronic devices and the rise of microelectronics.

Guglielmo Marconi (1874-1937) – A pioneer in wireless communication, Marconi developed the first practical radio transmission system, which became the foundation for modern telecommunications.

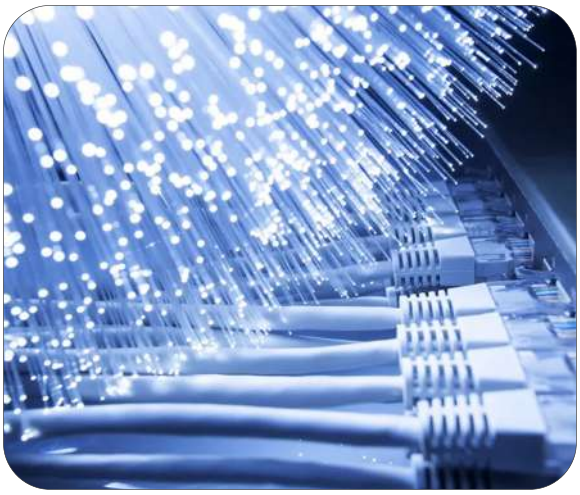
Tim Berners-Lee (Born 1955) – The inventor of the World Wide Web, Berners-Lee revolutionized communication by enabling the seamless exchange of information over the internet.

Shuji Nakamura (Born 1954) – Awarded the Nobel Prize for his invention of blue LEDs, which led to energy-efficient lighting and advancements in display technology.

Charles Kao (1933-2018) – Known as the "Father of Fiber Optics," Kao's research on optical fiber communication revolutionized high-speed internet and global connectivity.

Amar Bose (1929-2013) – Founder of Bose Corporation, he contributed significantly to audio engineering and sound systems, enhancing modern acoustics and communication technologies.

These pioneers have laid the foundation for numerous innovations in electronics and communication, shaping the world we live in today. Their work continues to inspire engineers and researchers to push the boundaries of technology.



Student Achievements and Awards

R Hariprasad Recognized as a Winner of Allianz Tech Championship 2024

Announcing that R Hariprasad of S6 ECE , representing Government Engineering College, Barton Hill, Thiruvananthapuram, has been recognized as one of the winners of the Allianz Tech Championship 2024, organized by Allianz Technology.



Kedar Kamalraj Receives IEEE Kerala Section Outstanding Student Volunteer Award 2024

Kedar Kamalraj, a student of S6 Electronics and Communication Engineering, has been honored with the IEEE Kerala Section Outstanding Student Volunteer Award 2024 for his exceptional dedication, leadership, and contributions to the IEEE community.



Reshma Ramesh and Anamika A Kamath Receive 2024 IEEE MTT-S Outstanding Volunteer Award

Reshma Ramesh and Anamika A Kamath, students of S6 Electronics and Communication Engineering, have been honored with the prestigious 2024 IEEE Microwave Theory and Techniques Society (MTT-S) Outstanding Volunteer Award for their remarkable contributions to IEEE MTT-S activities.

The award was presented at MAPCON 2024, the flagship international conference on Microwave, Antennas, and Propagation, held in Hyderabad.



Akshay R S receives IEEE Education Society IEEE Student Leadership Award 2024

Akshay R S, a student(ECE'20-24), has been honored with the prestigious IEEE Education Society IEEE Student Leadership Award 2024 for his outstanding leadership and contributions to student engagement in educational initiatives.



Kavya S Nair, Malavika M M, and Berby Mia Joseph Awarded WIE Scholarship 2024-25

Kavya S Nair(S6 ECE), Malavika M M (ECE'20-24), and Berby Mia Joseph (ECE'20-24), students of S6 Electronics and Communication Engineering, have been awarded the prestigious WIE Scholarship 2024-25 for their outstanding contributions to IEEE Women in Engineering (WIE) initiatives and academic excellence.
participants.



Student Achievements and Awards

IEEE SB GECBH Honors Outstanding Volunteers at AGM 2025

At the IEEE SB GECBH Annual General Meeting 2025, three outstanding students were recognized for their exceptional contributions to IEEE activities.

Ankitha V N (S6 ECE) was awarded the IEEE SB GECBH Outstanding Student Volunteer Award 2024 for her remarkable dedication, leadership, and active involvement in IEEE initiatives.

Kavya S Nair (S6 ECE) received the IEEE SB GECBH Outstanding RAS Volunteer Award 2024 for her exceptional contributions to the IEEE Robotics and Automation Society (RAS) Chapter.

Niranjan B Nair (S4 ECE) was honored with the IEEE SB GECBH Outstanding Design Volunteer Award 2024 for his innovative and creative contributions to the chapter's design projects.

These prestigious awards were presented by Principal Dr. Shiny G in recognition of their hard work and commitment to advancing the IEEE SB GECBH community.







Best Engineer 10.0

The Best Engineer 10.0' competition organized by IEEE SB GECBH celebrates exceptional technical talent and innovative thinking. Bharat Raj from S1 ECE emerged winner,, who demonstrated an impressive ability to solve engineering challenges. The competition emphasizes creativity, problem-solving, and the application of engineering principles in real-world scenarios



Kick Off 5.0

Sreya P Babu (S1 ECE) emerges as the winner of 'Kick Off 5.0,' a flagship event by IEEE SB GECBH designed to introduce and engage first-year students with the world of technology and engineering. Her success reflects her enthusiasm and adaptability in navigating complex technical challenges, marking the beginning of a promising academic journey.



Pixel Perfect 3.0

Unleash Your Creativity, an innovative initiative by IEEE SB GECBH aimed at encouraging creativity and forward-thinking. With the theme 'Unleash Your Creativity: Design the Future,' the event invites participants to explore and present transformative ideas, promoting a culture of innovation and interdisciplinary collaboration. Niranjan B Nair of S3 ECE was proclaimed the winner of this event.

Student Achievements and Awards



Techstreak 3.0

Raif Iza Muhamed (S4 ECE) is recognized as the winner of Techstreak 3.0, an event hosted by the IEEE APS and IEEE MTTS SBCs under IEEE SB GECBH. This technical competition emphasizes innovation in electromagnetics, antennas, and microwave technologies. Raif's victory highlights his technical expertise and his ability to navigate complex engineering challenges.

